

CUSTOMER MILESTONE BRIEF

MS #26 - Multi-Workload Factory Qualification

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.26.0, 0.26.1
Date	2026-08-27 to 2026-08-28
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #26 - Multi-Workload Factory Qualification converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Expands factory evaluation beyond one demonstration workload and calibrates qualification across different modernization patterns.

Why this matters to a customer

Expands factory evaluation beyond one demonstration workload and calibrates qualification across different modernization patterns.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added trusted workload profiles and public calibration catalogs for INTCALC, POSTTRAN, CREAMTMT, and the mixed PL/I-COBOL-Db2 ACCTPL1 cell.
- Added workload-specific private gates and deterministic repairs covering 23 published mutations plus eight clean accept-unchanged cases with zero false acceptance.
- Added a content-addressed qualification manifest and safety-first aggregate receipt requiring at least two distinct sealed model runs per workload.
- Bound qualification to exact evaluation, factory-run, model-call, request-manifest, portfolio, approval, checkpoint, latency, token, retry, resume, and cost evidence.
- Made one critical false acceptance block promotion regardless of aggregate repair rate.

- Expanded the portfolio to four work cells with graph/dependency conflict detection, two parallel waves, high-risk approval barriers, and checkpointed recovery that does not repeat passed cells.
- Added POSIX and PowerShell verification/qualification entry points and versioned JSON schemas.
- Kept model qualification, production authorization, native z/OS equivalence, and general workload coverage as separate claims; no model is qualified by committed public calibration.
- Added a fail-closed bridge for the exact retained v0.12 live OpenAI evaluation archive.
- Pinned the external archive and evaluation identities without committing the archive, credentials, raw prompts, or raw model responses.
- Verified legacy content hashes, artifact references, event-ledger chaining, model-call provenance, workspace reconstruction, and original baseline/final gate decisions.
- Published a schema-validated historical receipt and made it visible in the model-evidence dashboard.
- Kept public calibration, legacy schema, missing independent sealing, current-manifest gaps, token policy, workload repetition, and portfolio approval as explicit qualification blockers.
- Added Bash and PowerShell import commands plus tamper, secret, relabelling, traversal, and manifest- drift regression tests.

How it differs from earlier work

MS #26 builds on MS #25 (Reproducible PL/I Build Attestation). The earlier milestone established its own bounded capability; this milestone adds multi-workload factory qualification while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use multi-workload factory qualification as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Expanded the portfolio to four work cells with graph/dependency conflict detection, two parallel waves, high-risk approval barriers, and checkpointed recovery that does not repeat passed cells.

Source of record

- CHANGELOG.md release section(s): 0.26.0, 0.26.1
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.